

Search-FilesForWord.ps1

PowerShell script - recursive substring search for a word/text fragment across files in folders and subfolders.

```
<#
```

```
Search-FilesForWord.ps1
```

```
Searches for a word/text fragment inside files, recursively through subfolders.  
The search is a plain substring match (not "whole word"), so it will also match  
the term when it appears inside another word or combined with hyphens/underscores,  
e.g. searching for "server" will also match "fileserver", "server-01", "my_server", etc.
```

```
USAGE EXAMPLES:
```

```
# Search for "server-01" under the current folder, in all files
```

```
.\Search-FilesForWord.ps1 -SearchTerm "server-01"
```

```
# Search under a specific folder
```

```
.\Search-FilesForWord.ps1 -SearchTerm "server-01" -Path "C:\Logs"
```

```
# Limit the search to specific file types
```

```
.\Search-FilesForWord.ps1 -SearchTerm "server-01" -Path "C:\Logs" -Include *.log,*.txt,*.csv
```

```
# Case-sensitive search
```

```
.\Search-FilesForWord.ps1 -SearchTerm "Server-01" -CaseSensitive
```

```
#>
```

```
[CmdletBinding()]
```

```
param(
```

```
# The word / text fragment to search for. Matches as a substring, so hyphens and
```

```
# partial-word matches (e.g. inside "fileserver-01") are found as-is.
```

```
[Parameter(Mandatory = $true)]
```

```
[string]$SearchTerm,
```

```
# Root folder to search from. Defaults to the current directory.
```

```
[string]$Path = ".",
```

```
# Optional file name filters (wildcards), e.g. *.txt, *.log, *.csv.
```

```
# Leave empty to search every file.
```

```
[string[]]$Include = @(""),
```

```
# Perform a case-sensitive search. By default the search is case-insensitive.
```

```
[switch]$CaseSensitive
```

```
)
```

```
if (-not (Test-Path -LiteralPath $Path)) {
```

```
Write-Error "Path not found: $Path"
```

```
return
```

```
}
```

```
function Get-OccurrenceCount {
```

```
param(
```

```
[string]$Text,
```

```
[string]$Term,
```

```
[switch]$CaseSensitive
```

```
)
```

```
$comparison = if ($CaseSensitive) { [System.StringComparison]::Ordinal } else { [System.StringComparison]::OrdinalIgnoreCase }
```

```
$count = 0
```

```
$index = 0
```

```
while (($index = $Text.IndexOf($Term, $index, $comparison)) -ge 0) {
```

```
$count++
```

```

        $index += $Term.Length
    }

    return $count
}

$files = Get-ChildItem -Path $Path -Include $Include -Recurse -File -ErrorAction SilentlyContinue

$results = foreach ($file in $files) {
    $selectStringParams = @{
        Path      = $file.FullName
        Pattern    = $SearchTerm
        SimpleMatch = $true
        ErrorAction = "SilentlyContinue"
    }
    if ($CaseSensitive) {
        $selectStringParams["CaseSensitive"] = $true
    }

    Select-String @selectStringParams
}

if (-not $results) {
    Write-Host "No matches found for '$SearchTerm' under '$Path'."
    return
}

# Print every matching line, from the file it was found in, with how many
# times the search term appears on that specific line.
$results |
    Select-Object @{Name = "File"; Expression = { $_.Path } },
        @{Name = "Line"; Expression = { $_.LineNumber } },
        @{Name = "Text"; Expression = { $_.Line.Trim() } },
        @{Name = "Occurrences"; Expression = { Get-OccurrenceCount -Text $_.Line -Term $SearchTerm -CaseSensitive:$CaseSensitive } },
    Format-Table -AutoSize -Wrap

# Count how many times the word was found overall, and per file.
$perFileCounts = $results | Group-Object Path | ForEach-Object {
    [PSCustomObject]@{
        File = $_.Name
        Count = ($_.Group | ForEach-Object { Get-OccurrenceCount -Text $_.Line -Term $SearchTerm -CaseSensitive:$CaseSensitive }).Count
    }
}

Write-Host "Occurrences per file:"
$perFileCounts | Format-Table -AutoSize

$totalOccurrences = ($perFileCounts | Measure-Object -Property Count -Sum).Sum
Write-Host "`nTotal matching lines: $($results.Count)"
Write-Host "Total occurrences of '$SearchTerm': $totalOccurrences"

```